

DSC 40B Discussion 6 Worksheet

Friday, May 8, 2026

Imagine you are a tutor, and you need to explain something to someone who has only a vague understanding of what's going on. Do your best to get the point across. **ALTERNATE**.

If you're not sure, ask your peer if they know how to explain it, then REPEAT the explanation back. Remember, the point of the exercise is to **vocalize** your thoughts.

If you are both unsure, make sure to come to/tutor for help! Do not use ChatGPT or any other LLM! The point of discussion is to prepare you for the exams.

Have one person in the pair open the Discussion 1 Gradescope assignment. After both you **and** your partner finish a problem **and explain it**, answer the corresponding multiple choice/short answer questions for both versions.. You will receive instant feedback on the questions, whether you are correct or not. If your answer is incorrect, please go back to the question to understand it and feel free to ask tutors about anything you are confused about. Once you have finished the entire worksheet or discussion is ending, turn in your assignment and add your partner to the submission.

Quick Icebreaker

Introduce yourself to your partner by sharing name, year, major, etc. What has been the biggest surprise this quarter? (Do not spend more than 2 minutes!)

Questions

Question 1:

First Person:

You are a city planner responsible for mapping out and building roads for cars. Your boss wants you to model out your roads in a graph. What kind of graph would you use to model it? Explain your reasoning.

- ☐ Undirected Graph
- ☐ Directed Graph

What would the vertices of your graph represent? Explain your reasoning.

Second Person:

Since you did so well planning out roads, your boss has promoted you to also be responsible for planning and building hiker trails for the local national park. Once again, your boss wants you to

model out your trails in a graph. What kind of graph would you use to model it? Explain your reasoning.

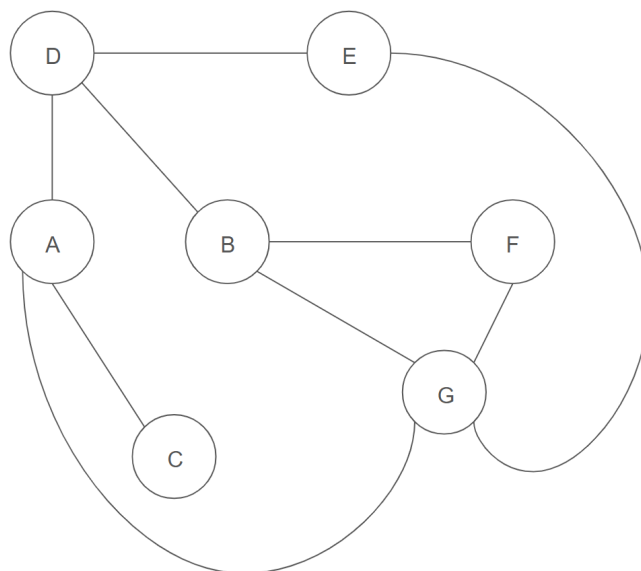
☐ Undirected Graph

☐ Directed Graph

What would the vertices of your graph represent? Explain your reasoning.

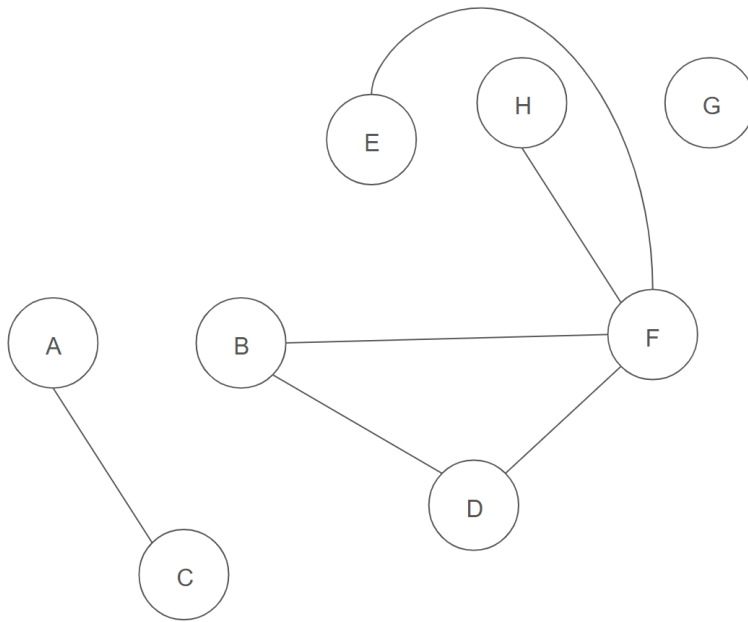
Question 2:

Second Person:



Given the graph above, list the set of vertices and edges.

First Person:



Given the graph above, list the set of vertices and edges.

Question 3:

First Person:

Given that there are 11 vertices in a directed graph, what is the maximum number of edges possible?

What if the graph is undirected?

Second Person:

Given that there are 14 vertices in a directed graph, what is the maximum number of edges possible?

What if the graph is undirected?

Question 4:

Second Person:

Given that there are 14 edges in a directed graph, what is the minimum and maximum number of vertices possible?

What if the graph is undirected?

First Person:

Given that there are 11 edges in a directed graph, what is the minimum and maximum number of vertices possible?

What if the graph is undirected?

Question 5:

First Person:

Given that an undirected graph has 8 vertices, what is the maximum degree a single node can have?

What if the graph is directed?

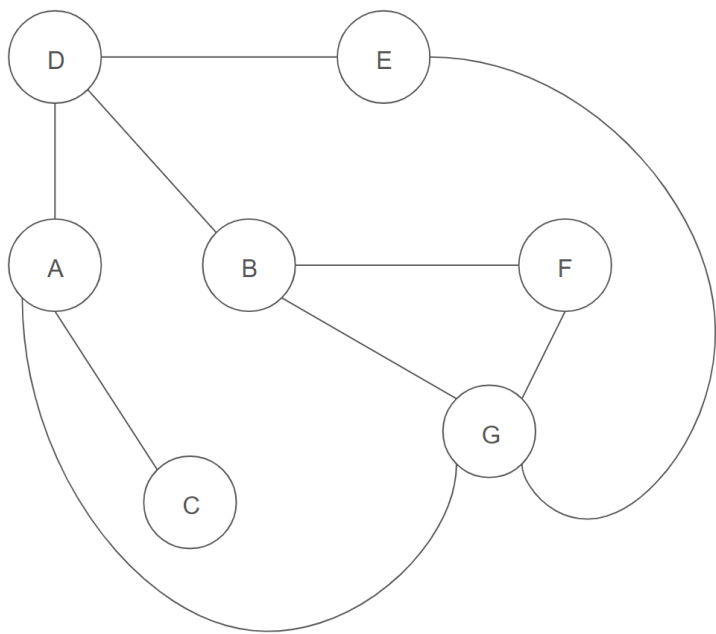
Second Person:

Given that an undirected graph has 9 vertices, what is the maximum degree a single node can have?

What if the graph is directed?

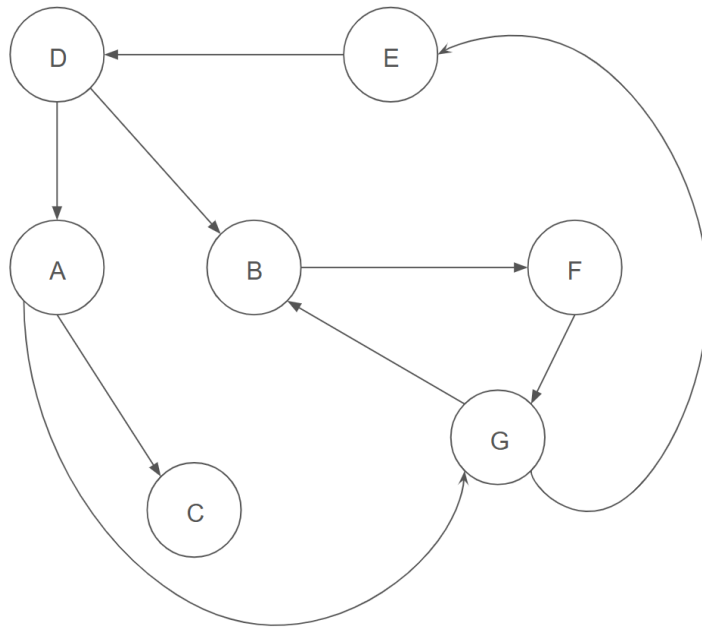
Question 6:

Second Person:



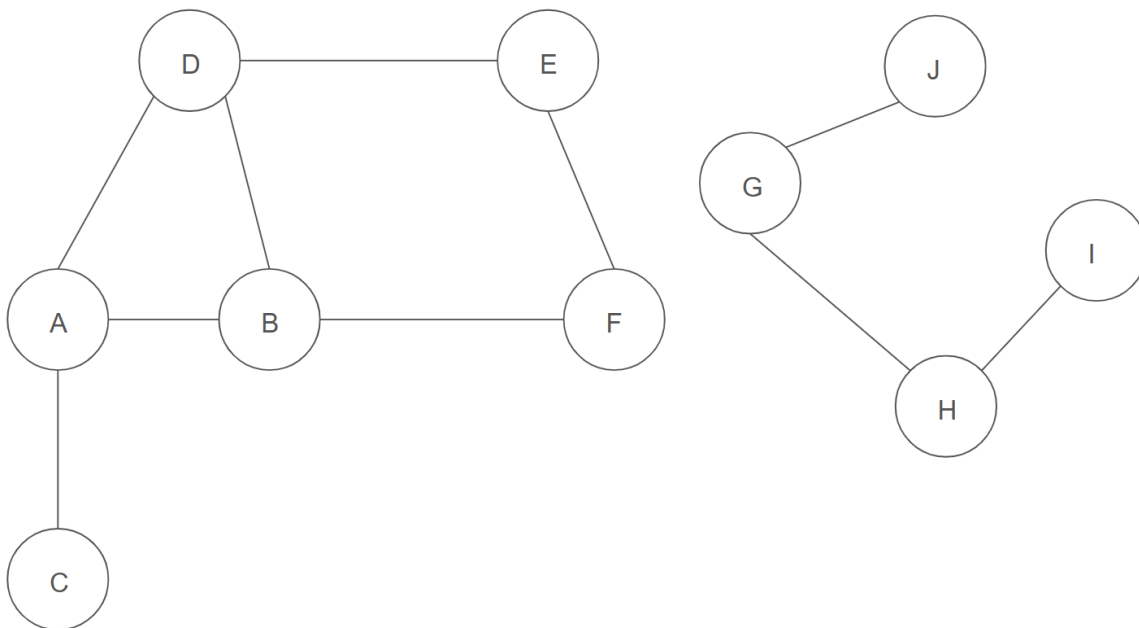
Give the graph above, list each node's neighbors.

First Person:



Given the graph above, list each node's neighbors.

Question 7:



First Person:

Name a path between vertex J and I.

What is the shortest path between vertex J and I?

Second Person:

Name a path between vertex A and F?

What is the shortest path between vertex A and F?

Question 8:

Second Person:

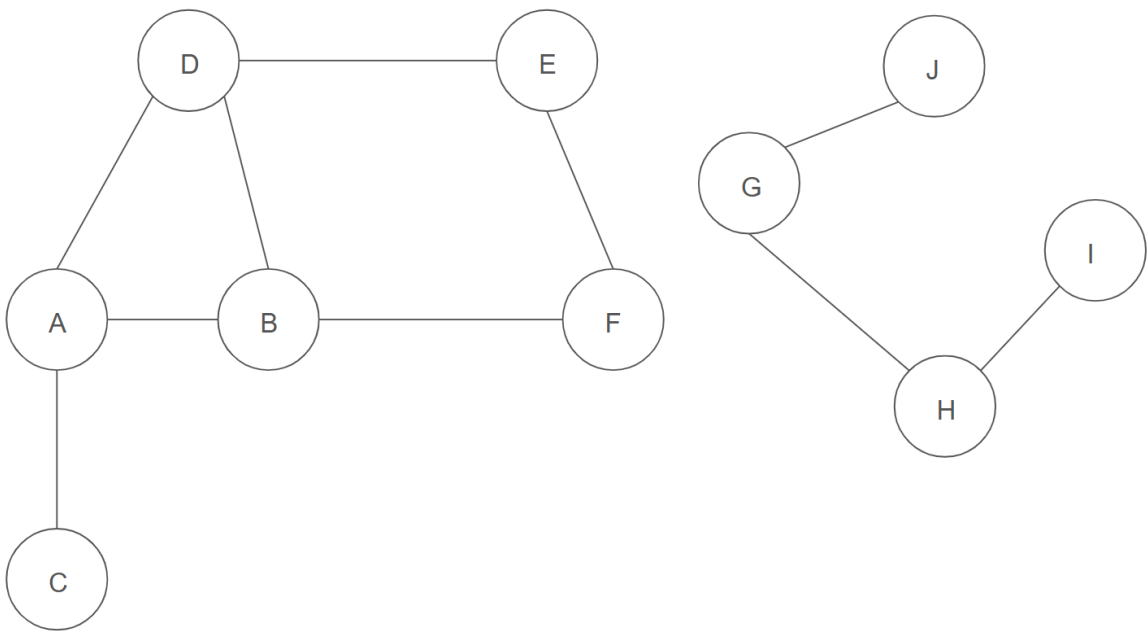
What is the longest path between vertex A and F?

First Person:

What is the longest path between vertex J and I?

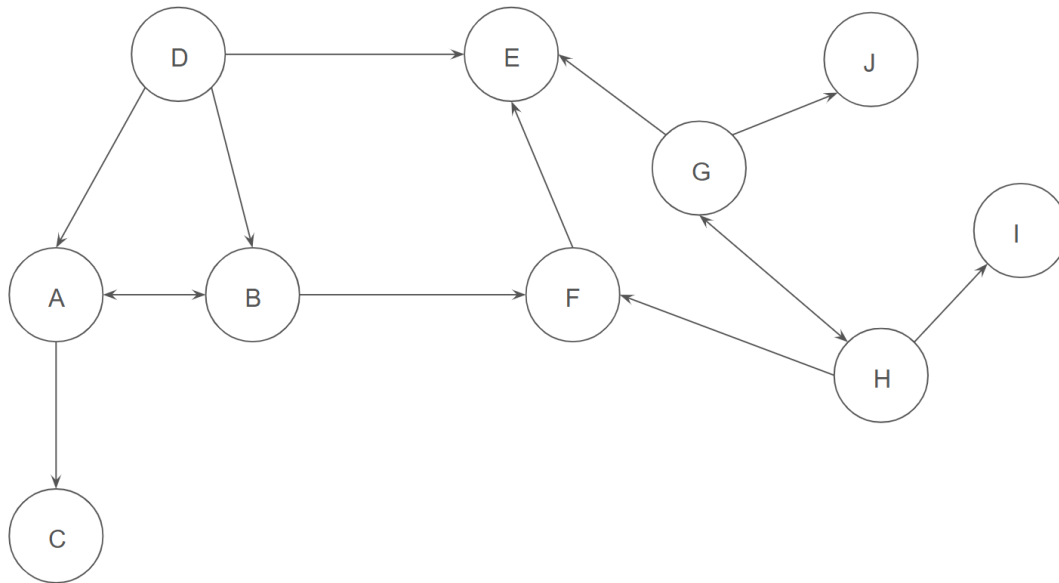
Question 9:

First Person:



Given the graph above, list all vertices reachable from vertex A.

Second Person:



Given the graph above, list all vertices reachable from vertex A.

Question 10:

Second Person:

For any connected **directed** graph, there exists a unique shortest path from each vertex to every other vertex in the graph.

- ☐ True
☐ False

First Person:

For any connected **undirected** graph, there exists a unique shortest path from each vertex to every other vertex in the graph.

- ☐ True
☐ False

Question 11:

First Person:

Given that an undirected graph has 8 vertices and 12 edges, what is the minimum and maximum number of connected components the graph can have?

Second Person:

Given that an undirected graph has 12 vertices and 8 edges, what is the minimum and maximum number of connected components the graph can have?

Question 12:**Second Person:**

If a graph is a binary tree, then its number of vertices is guaranteed to equal the number of edges plus one.

- ☐ True
- ☐ False

First Person:

If a graph's number of vertices is equal to the number of edges plus one, then it is guaranteed to be a binary tree.

- ☐ True
- ☐ False

Question 13:**First Person:**

Consider an undirected graph with 5 nodes. Which two of the following choices represent the same graph?

A.

```
adj = [  
    [1, 2],  
    [0, 3],  
    [0, 3, 4],  
    [1, 2],  
    [2]  
]
```

B.

```
adj = [  
    [1, 2],  
    [0, 2, 3],  
    [0, 1, 4],  
]
```



```
[1],  
[2]  
]
```

C.

```
adj = [  
    [1],  
    [0, 3],  
    [0, 3, 4],  
    [1, 2],  
    [2]  
]
```

D.

```
adj_matrix = [  
[0, 1, 1, 0, 0],  
[1, 0, 0, 1, 0],  
[1, 0, 0, 0, 1],  
[0, 1, 0, 0, 0],  
[0, 0, 1, 0, 0]  
]
```

E.

```
adj_matrix = [  
    [0, 1, 1, 0, 0],  
    [1, 0, 1, 1, 0],  
    [1, 1, 0, 0, 1],  
    [0, 1, 0, 0, 0],  
    [0, 0, 1, 0, 0]  
]
```

F.

```
adj_matrix = [  
    [0, 1, 1, 0, 0],  
    [1, 0, 0, 1, 0],  
    [1, 0, 0, 1, 1],  
    [0, 1, 1, 0, 0],  
    [0, 0, 1, 0, 0]  
]
```

Second Person:

Consider a directed graph with 5 nodes. Which two of the following choices represent the same graph?

A.

```
adj = [  
    [1],  
    [0, 3],  
    [0, 3, 4],  
    [1, 2],  
    [2]  
]
```

```
[1, 3],  
[2],  
[4],  
[2],  
[3]  
]
```

B.

```
adj = [  
    [1],  
    [2, 3],  
    [0, 4],  
    [2],  
    [3]  
]
```

C.

```
adj = [  
    [1, 3],  
    [2],  
    [0, 4],  
    [2],  
    [3]  
]
```

D.

```
adj_matrix = [  
    [0, 1, 0, 1, 0],  
    [0, 0, 1, 0, 0],  
    [1, 0, 0, 0, 1],  
    [0, 0, 1, 0, 0],  
    [0, 0, 0, 1, 0]  
]
```

E.

```
adj_matrix = [  
    [0, 1, 0, 1, 0],  
    [0, 0, 1, 0, 0],  
    [1, 0, 0, 0, 1],  
    [0, 1, 0, 0, 0],  
    [0, 0, 0, 1, 0]  
]
```

F.

```
adj_matrix = [  
    [0, 1, 0, 1, 0],  
    [0, 0, 1, 0, 0],  
]
```

```
[0, 0, 0, 0, 1],  
[0, 0, 1, 0, 0],  
[0, 0, 0, 1, 0]  
]
```

Question 14:

You are running a large scale social network app with millions of users where users add each other as friends.

Second Person:

You want to check if two people are friends or not. Which graph implementation of your social network would have better runtime efficiency?

- ☐ Adjacency List
- ☐ Adjacency Matrix

First Person:

You want to check how many friends a specific person has. Which graph implementation of your social network would have better runtime efficiency?

- ☐ Adjacency List
- ☐ Adjacency Matrix